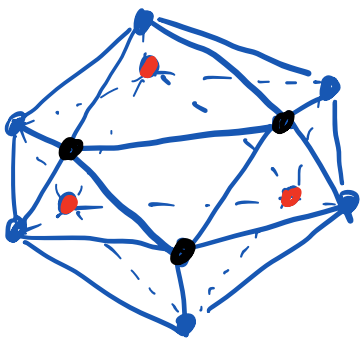
5 种 regular polyhedron

4, 6, 8, 12, 20

↓

C_{60}



icosahedron

20 面体的 五沿

5

10 (waist)

5

(20 triangles)

1 蓝色 6 个 (实线)

2 红色 3 个 (虚线)

3 黑色 3 个 (实线)

正 12 面体: 是 12 个正五边形



dodecahedron

$3 \times 108° < 360°$

足球：12个正5边形

20个正6边形

$2 \times \frac{5}{6} = 3/5$

每个正五边形的边跟一个正六边形

共 $12+20 = 32$ faces

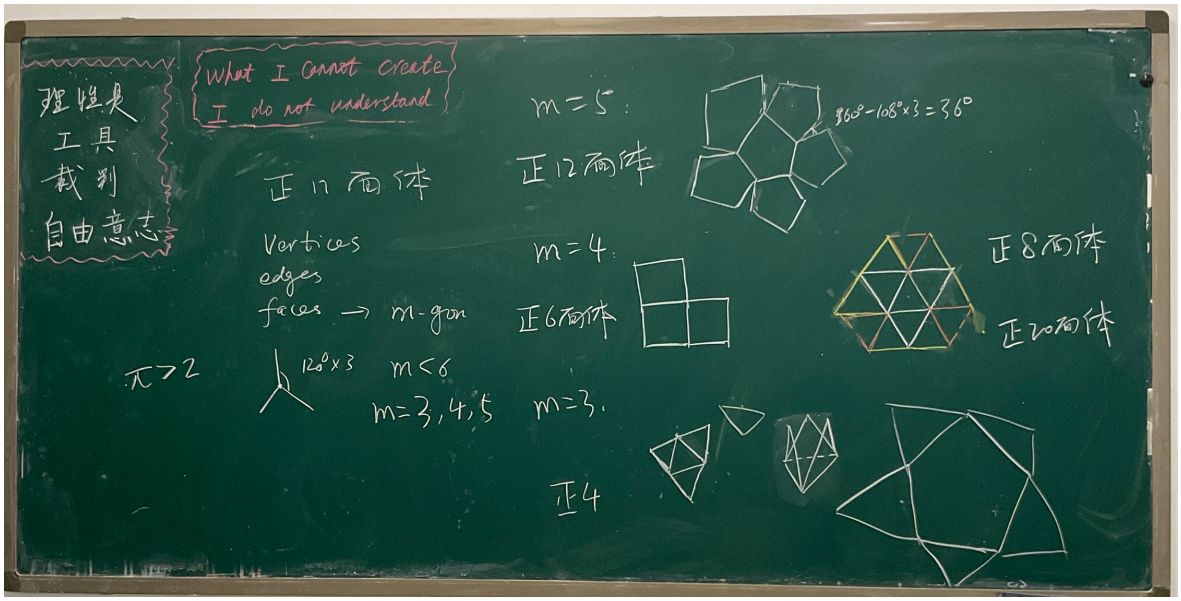
每条边被两个正多边形共用

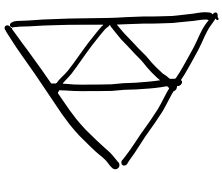
$(12 \times 5 + 20 \times 6)/2 = 90$ edges

每顶点被三个正多边形共用

$(12 \times 5 + 20 \times 6)/3 = 60$ vertices

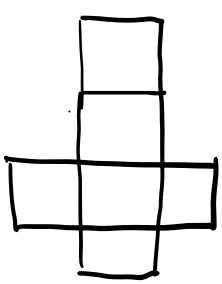
$F + V - E = 2$ Euler theorem





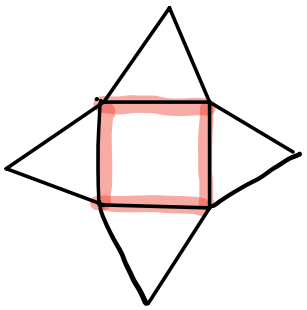
tetrahedron

正四面体



hexahedron

正六面体



two- layers

octahedron

正八面体

用一个顶点的平面铺排与环之差计算顶点个数：

球：　2个正六边形 + 1个正五边形

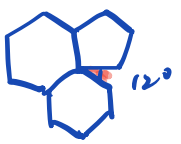
Solid angle

$4\pi = 720°$

↓

$4\pi r^2 / r^2$

总曲率分配到顶点上



$\delta = 12°$

angle defect

（角亏）

$720° / 12° = 60$ 个 vertices

正十二面体：　三个五边形



$360° - 108° \times 3 = 36°$

$720° / 36° = 20$ 个顶点

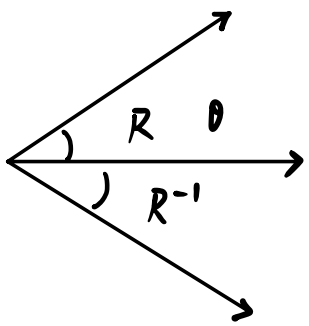
Gauss - Bonnet theorem

对于任何拓扑等价于球面的多面体（凸多面体都是），所有顶点的角亏之和是一个常数，且等于4π。 $\sum_{i=1}^{V} \delta_i = 4\pi$

Rotation : C_n

$$\vec{r} = m\vec{a} + n\vec{b}$$

$$m, n \in \mathbb{Z}$$



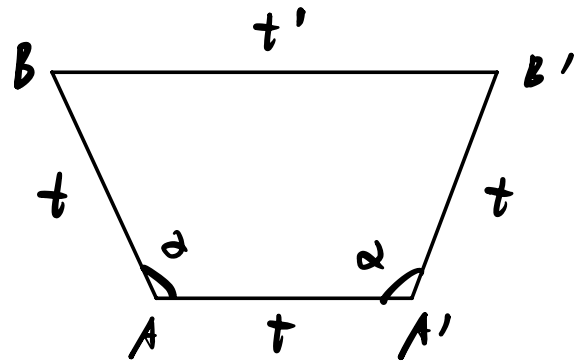
$$R\vec{r} + R^{-1}\vec{r} = p\vec{r}$$

$$2\cos\theta\,\vec{r} = p\vec{r}$$

$$p \in \mathbb{Z}, \quad 2\cos\theta = p$$

$$-1 \le \cos\theta \le 1$$

Another way :



$R \, 5 \, R^{-1}$

$A' - A = t$

$t' = mt$ (crystal)

$t' = xt \sin(\alpha - 90°) + t$

$= -2 + \cos\alpha + t \quad = mt$

$m \in \mathbb{Z}$

$\cos\alpha = (1-m)/2 \quad \in [-1, 1]$

$m = 3, 2, 1, 0, -1$

$\alpha = \pi, \frac{2\pi}{3}, \frac{\pi}{2}, \frac{\pi}{3}, 0$

$C_2, \quad C_3, \quad C_4, \quad C_6, \quad E \ (C_1)$

$p=1$, $\theta=60°$ C_6

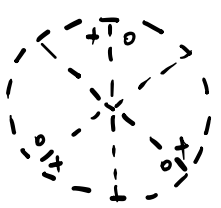
$p=-1$, $\theta=120°$, C_3

$p=0$, $\theta=90°$, C_4

$p=2$, $\theta=0$, E

$p=-2$, $\theta=180°$, C_2

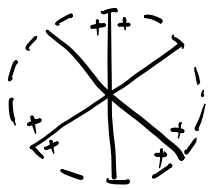
D_3 和 C_{3v}



D_3

包含一个 n 度主轴 和

n 个与之垂直的 2 度轴



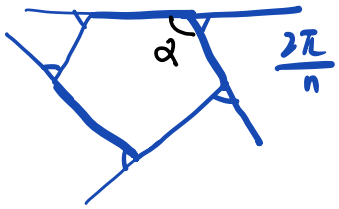
C_{3v}

含有 n 度主轴 和

过主轴的 n 个垂直镜面

Platonic solids

正多面体，V, E, F 节约

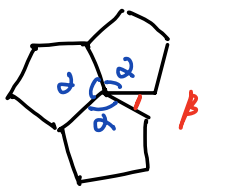




m个正n边形拼接

$(\pi - \frac{2\pi}{n})$ 为 α

$m\alpha < 2\pi$



$\beta > 0$

$3\alpha < 2\pi$

$n \geq 3$ （没有二边形）

$m = 3, 4, 5$，$\times$

$n=5$, $m=3$, 正12

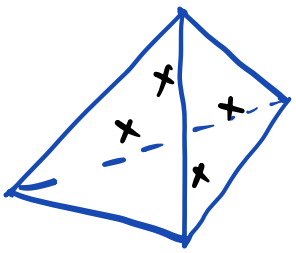
$n=4$, $m=3$, 正6

$n=3$, $m=3$, 正4

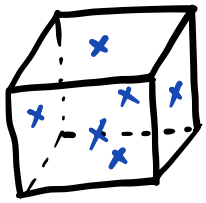
$m=4$, 正8

$m=5$, 正20

Tetrahedron



4个面上4个中心为顶组成正四方体，自对偶



Cube 6个面上中心为组成正8面体（6个顶点）

反过来，正8面体 8个面心构成Cube的8个顶点

正6面体与正8面体互相对偶

分析：正12面体

12 faces

$12 \times 5 = 60$，$60/2 = 30$ edges

1个边需要2个顶点，1个顶点连3条边

共 $30 \times 2/3 = 20$ vertices

$$\frac{720}{360-(3\times 10)} = 20$$

正20面体

20 faces

$20 \times 3 = 60$，$60/2 = 30$ edges

1个边需要2个顶点，1个顶点连5条边

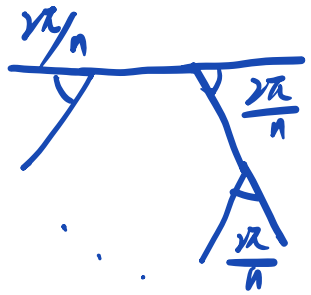
$30 \times 2/5 = 12$ vertices

$$\frac{720}{360-(5\times 60)} = 12$$

正12面体与正20面体对偶。

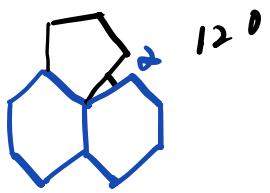
"补角":

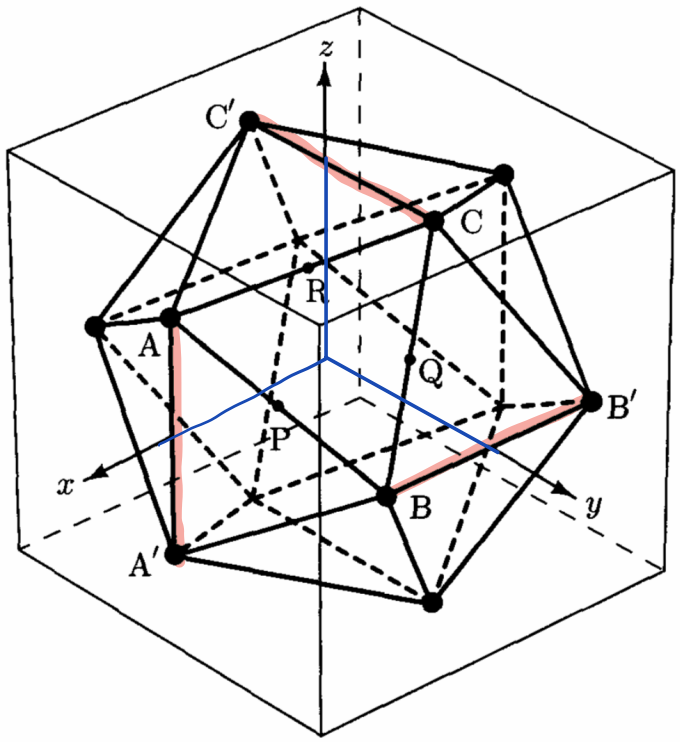
平面: $2\pi / \frac{2\pi}{n}$ = n 个顶点



球面 $4\pi / \alpha$ = 60 vertices

足球:





$\cos\frac{\pi}{5} = \frac{1}{\tau}$

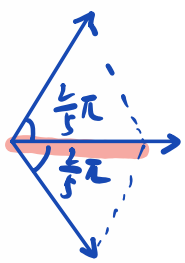


Fig. 2.10.1. An icosahedron showing its 12 vertices, 20 triangular faces, and 30 edges. The five-fold axes pass through the vertices, the three-fold axes through the centers of the faces, and the two-fold axes through the centers of the edges. The icosahedron with edges of unit length is inscribed in a cube. Its edges AA′, BB′, and CC′ are, respectively, parallel to the z-, x-, and, y-axes of the cube. The coordinates of A, A′, B, B′, C, and C′ are, respectively, $(\tau, 0, \pm 1)$, $(\pm 1, \tau, 0)$, and $(0, \pm 1, \tau)$, where $\tau = (1 + \sqrt{5})/2$ is the golden mean. The coordinates of P, Q and R are, respectively, $(1 + \tau, \tau, 1)/2$, $(1, 1 + \tau, \tau)/2$, and $(\tau, 1, 1 + \tau)/2$.

$\tau^2 = \tau + 1$

正20面体与正6面体为.15 ?

$a_6/a_{20} = \sqrt{\frac{1}{4} + \frac{\sqrt{5}}{2}} \approx 1.1175$

对顶点： 6对， 6个 five-fold

对面： 10对， 10个 three-fold

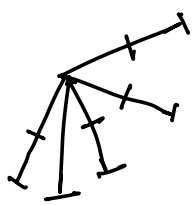
对边： 15对 15个 two-fold

dodecahedron 正12面体

还可以在这里讲置换群，比如正五边形组成的正多面（12面）体， 20 vertices, 30 edges

观察一个顶点，有3个五边形. 所以有

C_3 , C_3^{-1} （轴线：穿过此顶点与体心）



有 10对这样的顶点

$10 \times 2 = 20$ 个对称操作

30 edges： 15对 边 C_2

15个 elements

6个对面，　　C_5

$6 \times 4 = 24$ 个操作

共有　e（单位元）+ 20 + 15 + 24 = 60

↓

（12 + 12）

$\pm\frac{2}{5}\pi$　$\pm\frac{4}{5}\pi$

个对称操作

Conjugate class of size : 1, 20, 15, 12, 12

见　Benedict gross　Lec 19

2003.10.31